

Lecture 2:

Smoothed Particle Hydrodynamics (SPH) Formulation

Professor G.R. Liu

SMA Fellow, Director, ACES Centre

Department of Mechanical Engineering, National University of Singapore

(Office: EA-5-05; Tel: 6516-6481; E-mail: mpeliugr@nus.edu.sg)

<http://www.nus.edu.sg/ACES/>

Collaborator: Dr. Li Zirui

Introduction to SPH

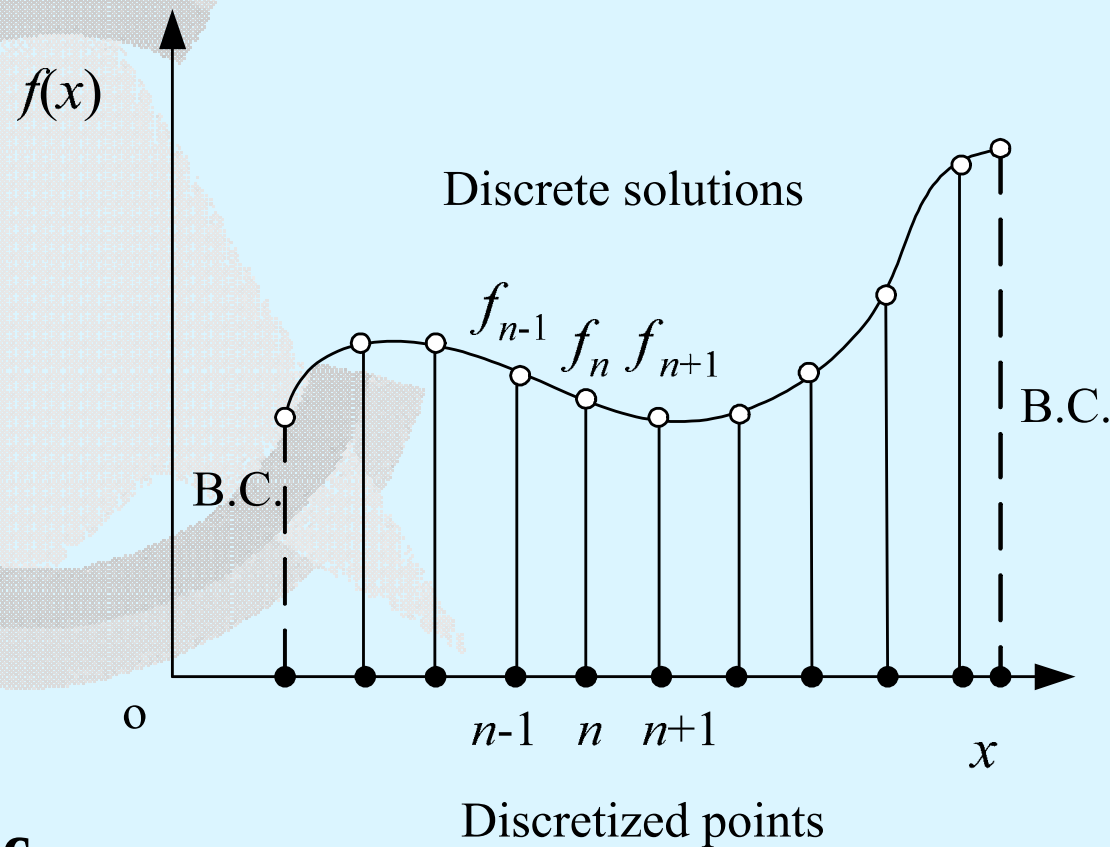
➤ Basic concepts of computational hydrodynamics

- **Grid based methods**
 - ✓ Eulerian approach (FDM)
 - ✓ Lagrangian approach (FEM)
 - ✓ Meshfree methods can use both approaches
- **Particle based methods**
 - ✓ Smoothed Particle Hydrodynamics
 - ✓ Dissipative Particle Dynamics
 - ✓ Brownian Dynamics

Introduction to SPH

- A typical numerical simulation of a CFD problem involves the following factors.

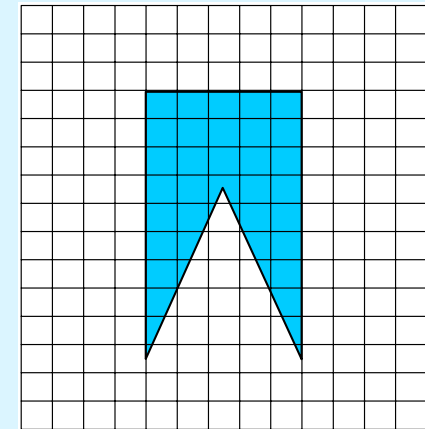
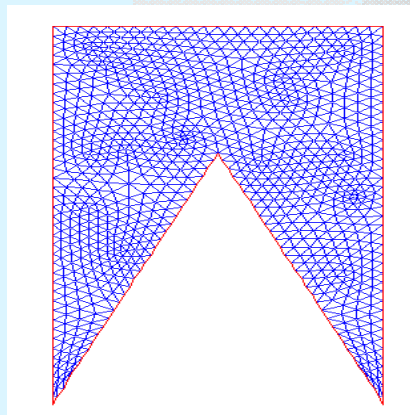
- ✓ Governing equations,
- ✓ Proper boundary conditions and/or initial conditions,
- ✓ Domain discretization,
- ✓ Formulation technique,
- ✓ Solving the resultant algebraic equations or ordinary differential equations (ODE).



Introduction to SPH: descriptions

Eulerian vs Lagrangian descriptions

	Lagrangian methods	Eulerian methods
Grid	Attached on the moving material	Fixed in the space
Track	Movement of any point on materials	Mass, momentum, and energy flux across grid nodes and mesh cell boundary
Time history	Easy to obtain time-history data at a point attached on materials	Difficult to obtain time-history data at a point attached on materials
Moving boundary and interface	Easy to track	Difficult to track
Irregular geometry	Easy to model	Difficult to model with good accuracy
Large deformation	Difficult to handle	Easy to handle



Introduction to SPH

Meshfree particle methods (MPMs)

Methods	References
Molecular dynamics (MD)	Alder and Wainright ,1957; Rahman, 1964; Stillinger and Rahman, 1974; etc.
Monte Carlo (MC)	Metropolis and Ulam, 1949; Binder, 1988, 1992; etc.
Direct simulation Monte Carlo (DSMC)	Bird, 1994; Pan et al., 1999, 2000, 2002; etc.
Dissipative particle dynamics (DPD)	Hoogerbrugge and Koelman, 1992; Español, 1998; etc.
Lattice gas Cellular Automata (CA)	Wolfram, 1983; Kandanoft et al. 1989; etc.
Lattice Boltzmann equation (LBE)	Chen and Doolen, 1998; Qian et al., 2000; etc.
Particle-in-Cell (PIC)	Harlow, 1963; 1964; etc.
Marker-and-Cell (MAC)	Harlow, 1964; etc.
Fluid-in-Cell (FLIC)	Gentry et al., 1966; etc.
Moving Particle Semi-implicit (MPS)	Koshizuka et al., 1998; etc.
Discrete element method (DEM)	Cundall, 1987; Owen, 1996; etc.
Vortex methods	Chorin, 1973; Leonard, 1980; etc.
Smoothed particle hydrodynamics (SPH)	Lucy, 1977; Gingold and Monaghan, 1977; etc.

SPH: derivatives

Grid based methods

- the Eulerian description is a spatial description, e.g. FDM
- the Lagrangian description is a material description, e.g. FEM

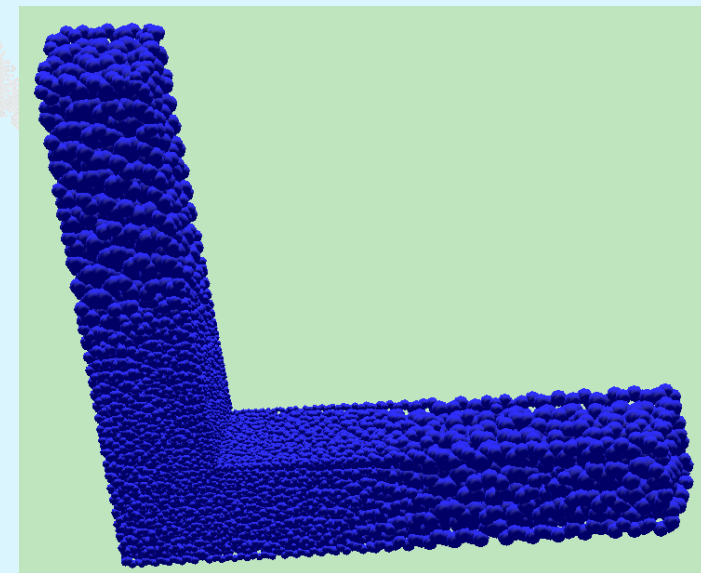
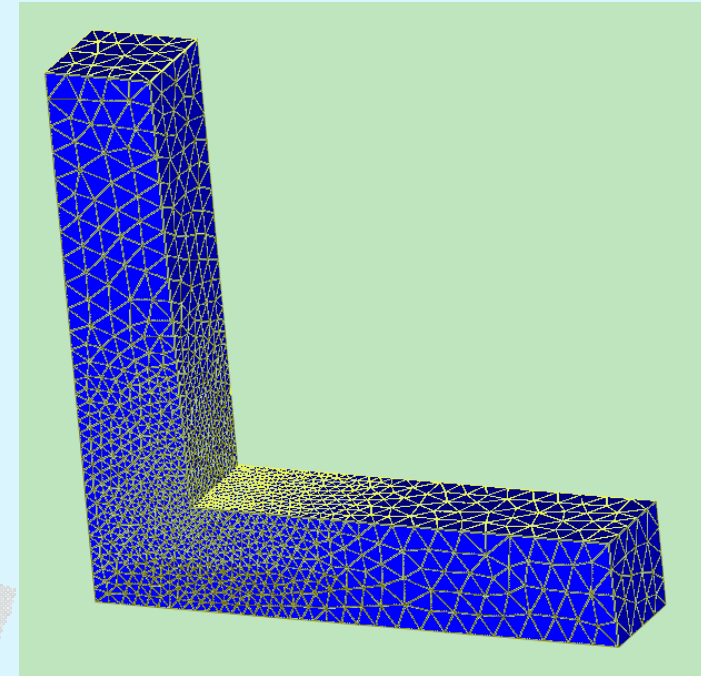
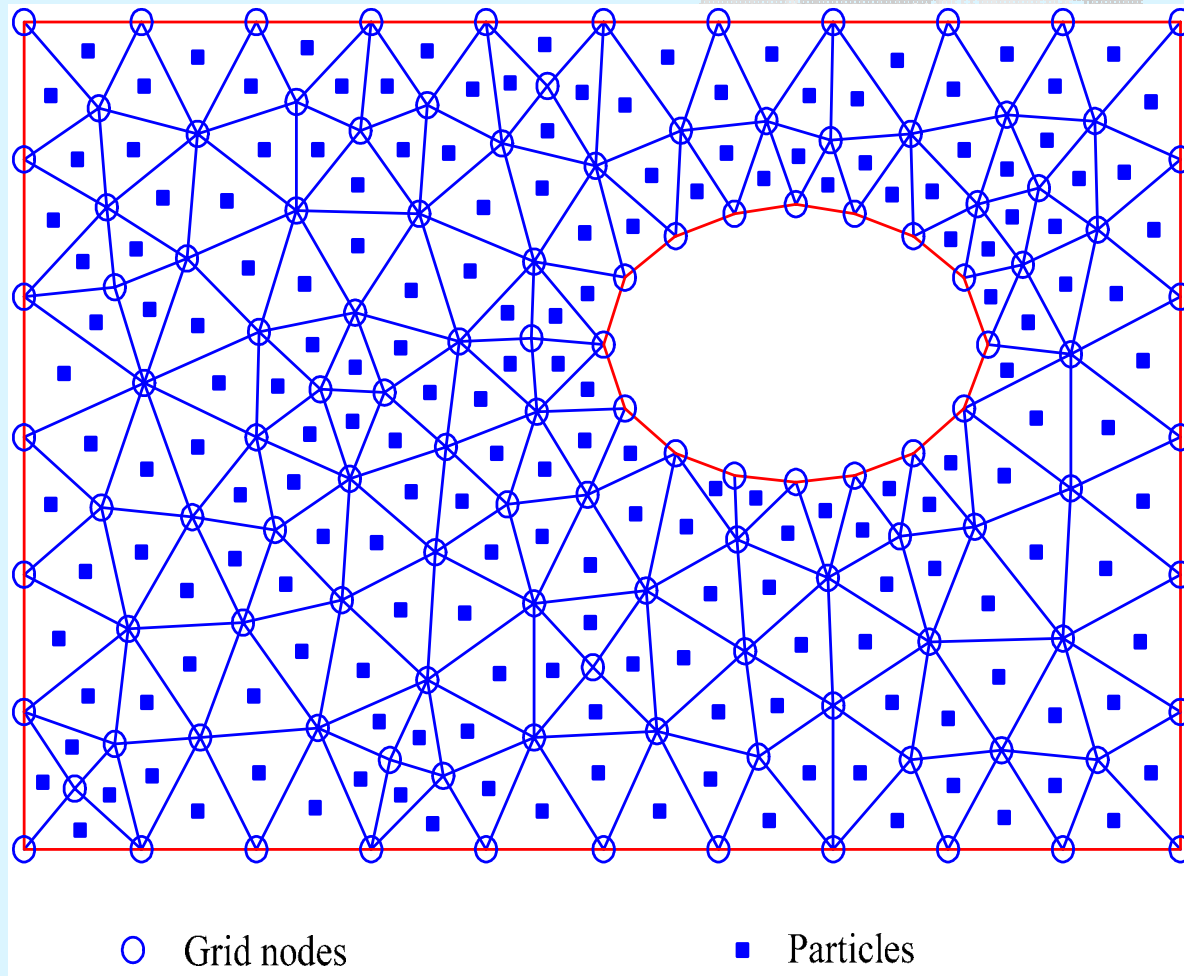
$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v}^\alpha \frac{\partial}{\partial \mathbf{x}^\alpha}$$

Conservation	Lagrangian description	Eulerian description
Mass	$\frac{D\rho}{Dt} = -\rho \frac{\partial \mathbf{v}^\beta}{\partial \mathbf{x}^\beta}$	$\frac{\partial \rho}{\partial t} + \mathbf{v}^\beta \frac{\partial \rho}{\partial \mathbf{x}^\beta} = -\rho \frac{\partial \mathbf{v}^\beta}{\partial \mathbf{x}^\beta}$
Momentum	$\frac{D\mathbf{v}^\beta}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \mathbf{x}^\beta}$	$\frac{\partial \mathbf{v}^\beta}{\partial t} + \mathbf{v}^\alpha \frac{\partial \mathbf{v}^\beta}{\partial \mathbf{x}^\alpha} = -\frac{1}{\rho} \frac{\partial p}{\partial \mathbf{x}^\beta}$
Energy	$\frac{De}{Dt} = -\frac{p}{\rho} \frac{\partial \mathbf{v}^\beta}{\partial \mathbf{x}^\beta}$	$\frac{\partial e}{\partial t} + \mathbf{v}^\beta \frac{\partial e}{\partial \mathbf{x}^\beta} = -\frac{p}{\rho} \frac{\partial \mathbf{v}^\beta}{\partial \mathbf{x}^\beta}$

N.S equations for inviscid, no heat transfer flows

SPH solution strategy

Initial particle discretization: use grids

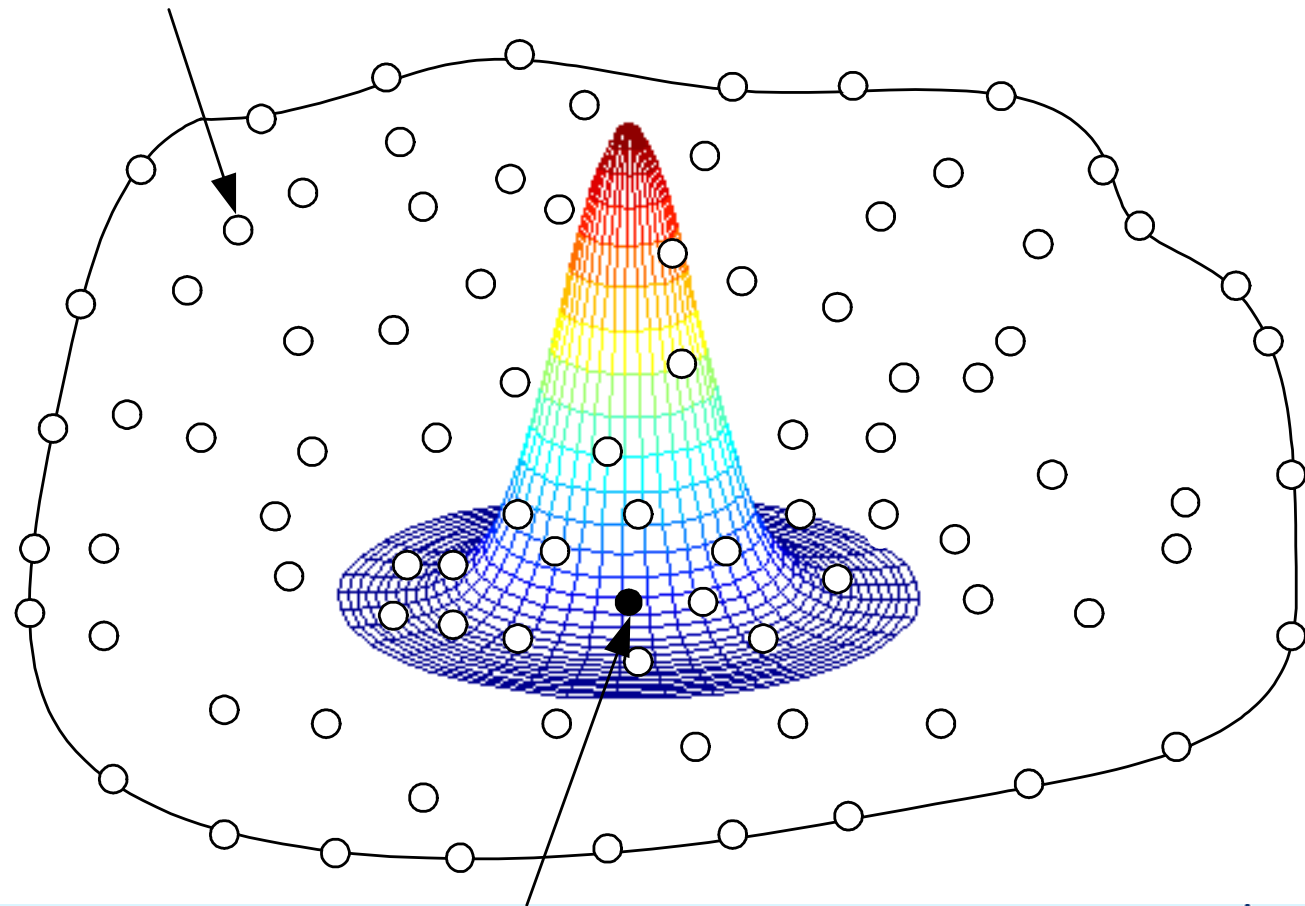


SPH: Particle approximation

To approximate the values of functions, derivatives at a particle using the information at all the neighboring particles

$$u(\mathbf{x}) = \sum_{i=1}^N \phi_i(\mathbf{x}) u_i$$

Meshfree particles



Particle at $\mathbf{x}$

SPH Formulation

➤ Basic concepts (particle and explicit)

1. Discretization using a set of arbitrarily distributed *particles*.
2. Integral function approximation: kernel approximation
3. Particle approximation of field functions.
 - ✓ Summation to replace integration
 - ✓ Field function and its derivatives
4. PDEs are represented directly in particle approximation.
5. No connectivity is defined between particles: *large deformation*.
6. The ODE's are solved using explicit integration algorithm: *explicit*.

SPH Formulation: function approximation

Integral representation of a function

- Starting form
$$f(\mathbf{x}) = \int_{\Omega} f(\mathbf{x}') \delta(\mathbf{x} - \mathbf{x}') d\mathbf{x}'$$

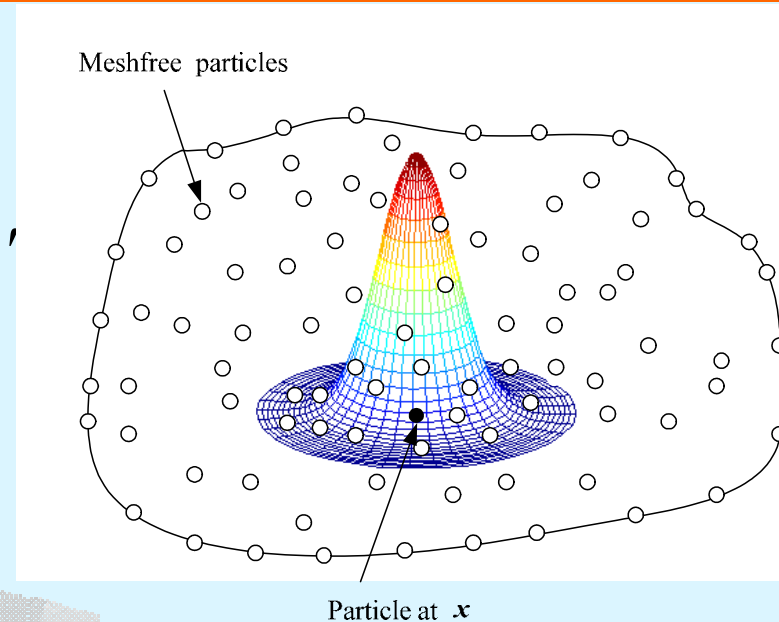
$$\delta(\mathbf{x} - \mathbf{x}') = \begin{cases} 1 & \mathbf{x} = \mathbf{x}' \\ 0 & \mathbf{x} \neq \mathbf{x}' \end{cases}$$

- Replace $\delta(\mathbf{x} - \mathbf{x}')$ by a smoothing function $W(\mathbf{x} - \mathbf{x}', h)$

$$f(\mathbf{x}) \approx \int_{\Omega} f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

- W is the so-called *smoothing kernel function*, or *smoothing function*, or *smoothing kernel*, or *kernel function* or *kernel*

$$\langle f(\mathbf{x}) \rangle = \int_{\Omega} f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$



SPH Formulation: function approximation

➤ Basic properties of smoothing function

- Normalization condition (unity condition)

$$\int_{\Omega} W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}' = 1$$

- Delta function property

$$\lim_{h \rightarrow 0} W(\mathbf{x} - \mathbf{x}', h) = \delta(\mathbf{x} - \mathbf{x}')$$

- Compact condition

$$W(\mathbf{x} - \mathbf{x}', h) = 0 \quad \forall |\mathbf{x} - \mathbf{x}'| > \kappa h$$

κ is a constant related to the smoothing function for point at $\mathbf{x}$, and defines the effective (non-zero) area of the smoothing function.

SPH Formulation: function approximation

➤ Derivative of a function

$$\langle \nabla f(\mathbf{x}) \rangle = \int_{\Omega} [\nabla f(\mathbf{x}')] W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

$$[\nabla f(\mathbf{x}')] W(\mathbf{x} - \mathbf{x}', h) =$$

$$\nabla [f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h)] - f(\mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}', h)$$

$$\langle \nabla \cdot f(\mathbf{x}) \rangle =$$

$$\int_{\Omega} \nabla [f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h)] d\mathbf{x}' - \int_{\Omega} f(\mathbf{x}') \nabla W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

$$\langle \nabla \cdot f(\mathbf{x}) \rangle = \int_S \underbrace{f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h) \cdot \vec{n}}_{=0, \text{ on } S} dS - \int_{\Omega} f(\mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

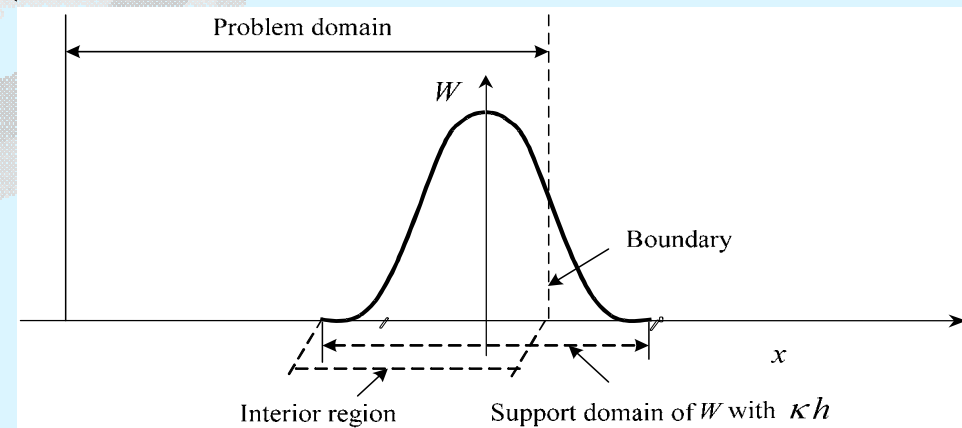
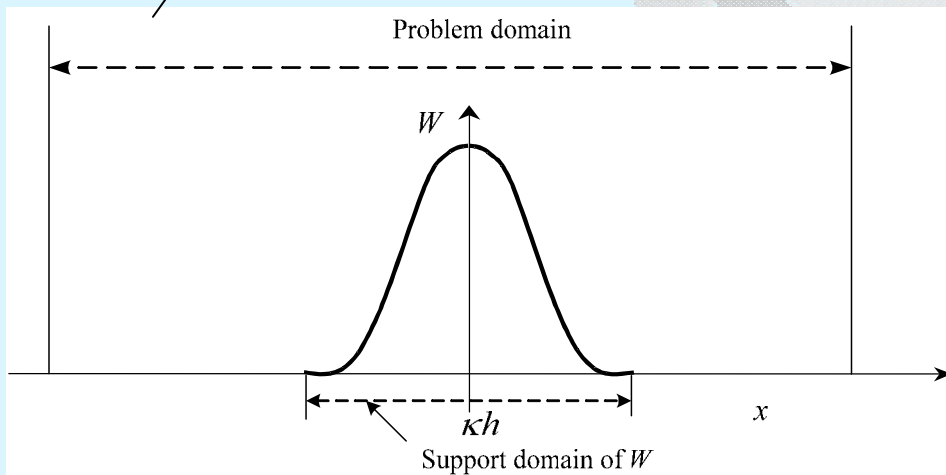
$$\langle \nabla \cdot f(\mathbf{x}) \rangle = - \int_{\Omega} f(\mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

SPH Formulation: function approximation

► Boundary effects

$$\langle \nabla \cdot f(\mathbf{x}) \rangle = - \int_{\Omega} f(\mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$

$$\langle \nabla \cdot f(\mathbf{x}) \rangle = \int_S f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h) \cdot \vec{n} dS - \int_{\Omega} f(\mathbf{x}') \cdot \nabla W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}'$$



Special treatment required

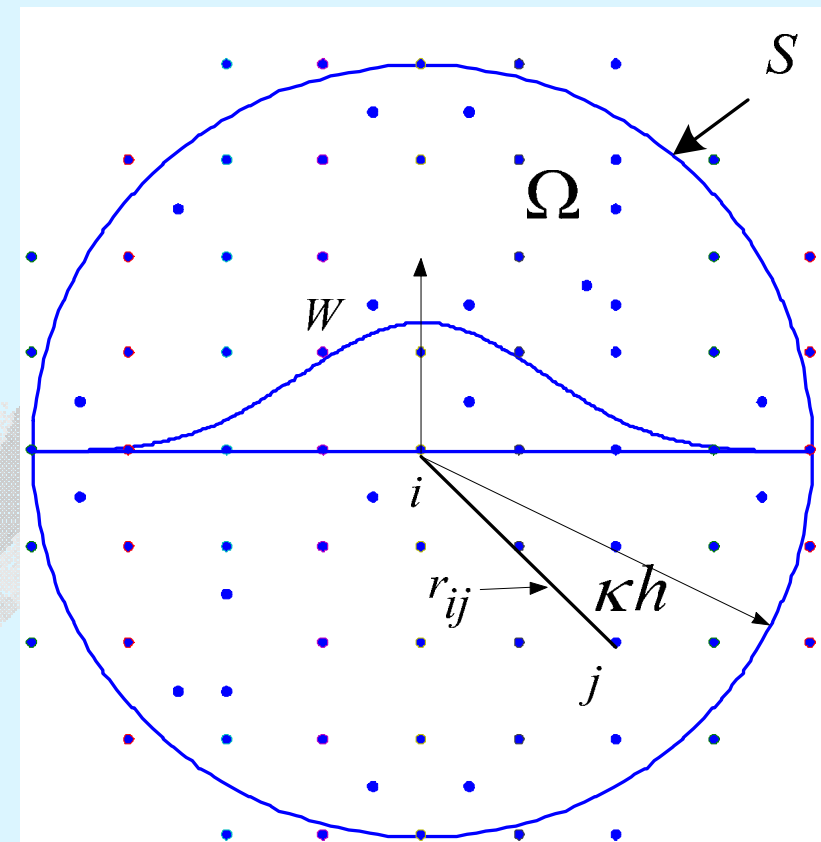
SPH Formulation: particle approximation

- The problem domain is represented by a finite number of particles that carry individual mass and occupy individual volume.
- Mass of a particle $m_j = \Delta V_j \rho_j$
- Particle approximation of $f(\mathbf{x})$

$$\begin{aligned}
 f(\mathbf{x}) &= \int_{\Omega} f(\mathbf{x}') W(\mathbf{x} - \mathbf{x}', h) d\mathbf{x}' \\
 &\cong \sum_{j=1}^N f(\mathbf{x}_j) W(\mathbf{x} - \mathbf{x}_j, h) \Delta V_j \\
 &= \sum_{j=1}^N f(\mathbf{x}_j) W(\mathbf{x} - \mathbf{x}_j, h) \frac{1}{\rho_j} (\rho_j \Delta V_j) \\
 &= \sum_{j=1}^N f(\mathbf{x}_j) W(\mathbf{x} - \mathbf{x}_j, h) \frac{1}{\rho_j} (m_j)
 \end{aligned}$$



$$\langle f(\mathbf{x}_i) \rangle = \sum_{j=1}^N \frac{m_j}{\rho_j} f(\mathbf{x}_j) \cdot W_{ij}$$

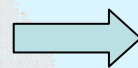


$$W_{ij} = W(\mathbf{x}_i - \mathbf{x}_j, h) = W(|\mathbf{x}_i - \mathbf{x}_j|, h)$$

SPH Formulation: particle approximation

➤ Function

$$\langle f(\mathbf{x}_i) \rangle = \sum_{j=1}^N \frac{m_j}{\rho_j} f(\mathbf{x}_j) \cdot W_{ij}$$



$$\rho_i = \sum_{j=1}^N m_j W_{ij}$$

➤ Derivative of a function

$$\langle \nabla \cdot f(\mathbf{x}) \rangle = - \sum_{j=1}^N \frac{m_j}{\rho_j} f(\mathbf{x}_j) \cdot \nabla W(\mathbf{x} - \mathbf{x}_j, h)$$

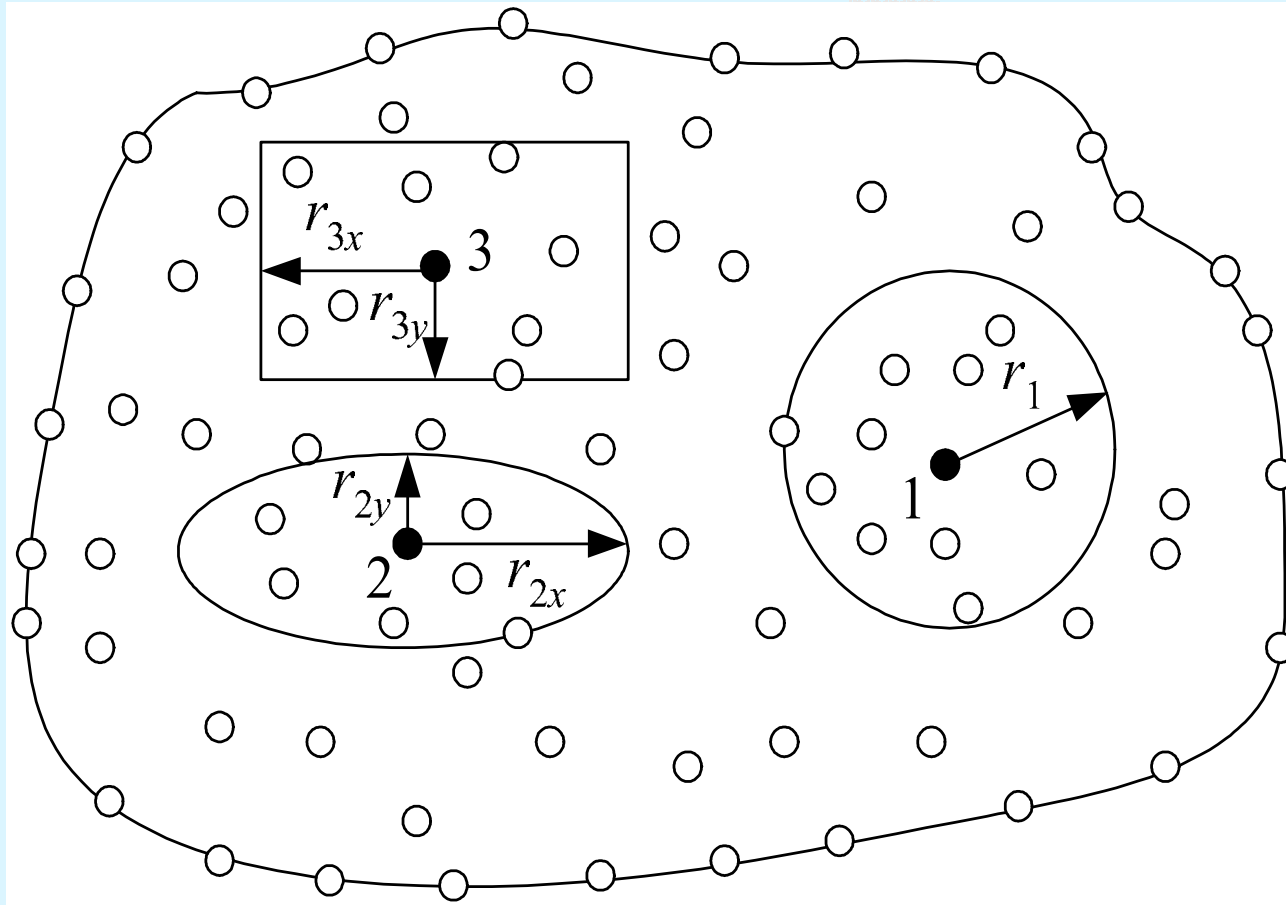
$$\langle \nabla \cdot f(\mathbf{x}_i) \rangle = - \sum_{j=1}^N \frac{m_j}{\rho_j} f(\mathbf{x}_j) \cdot \nabla_i W_{ij}$$

$$\nabla_i W_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{r_{ij}} \frac{\partial W_{ij}}{\partial r_{ij}} = \frac{\mathbf{x}_{ij}}{r_{ij}} \frac{\partial W_{ij}}{\partial r_{ij}}$$

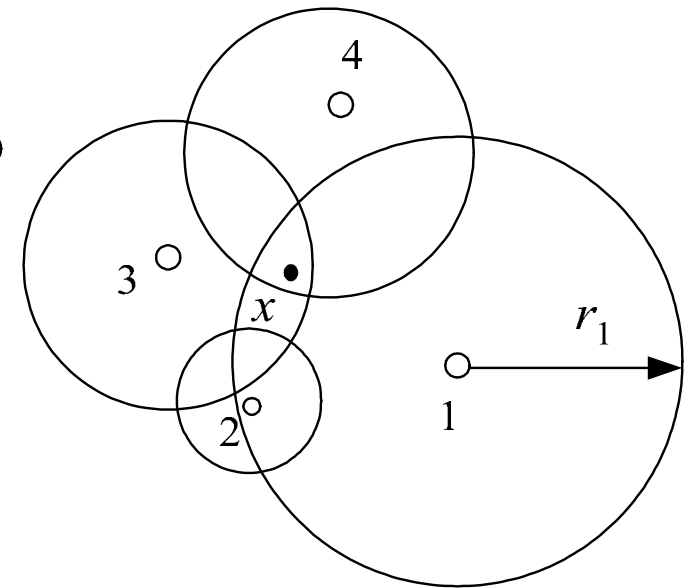
Particle selection: support & influence domains

- The support domain for a field point at $\mathbf{x}$ is the domain where the information for all the points inside this domain is used to determine the information at the point at $\mathbf{x}$.
- The influence domain is defined as a domain where a node exerts its influences.
- When the concept of support domain is used, the consideration is based on a field point; When the concept of influence domain is used, the consideration is based only on the nodes.
- If a node i is within the support domain of a point $\mathbf{x}$, then node i exert an influence on point $\mathbf{x}$, and thus point $\mathbf{x}$ is within the influence domain of node i .

Particle selection: support & influence domains



Support domain



Influence domain

SPH Formulation: concluding remarks

- The SPH method employs particles to represent material and form the computational frame. There is no need for predefined connectivity between these particles. All one needs is the initial particle distribution.
- The SPH approximation consists of kernel approximation and particle approximation. The kernel approximation of a function and its derivative are carried out in the continuum domain, and the particle approximations of a function and its derivative are carried out using discretized particles in the support domain.
- Each particle in the SPH method is associated with a support domain and influence domain. For most practical applications, the support domain of a particle can be equal to its influence domain.

SPH: references

- 1) Liu G.R. and Liu M.B., Smoothed Particle Hydrodynamics: a meshfree particle method, World Scientific, 1st printing (2003), 3rd printing (2007)
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- 4) Lucy L.B., Numerical approach to testing the fission hypothesis, Astronomical Journal, 82:1013-1024, (1977)
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Thank you!

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